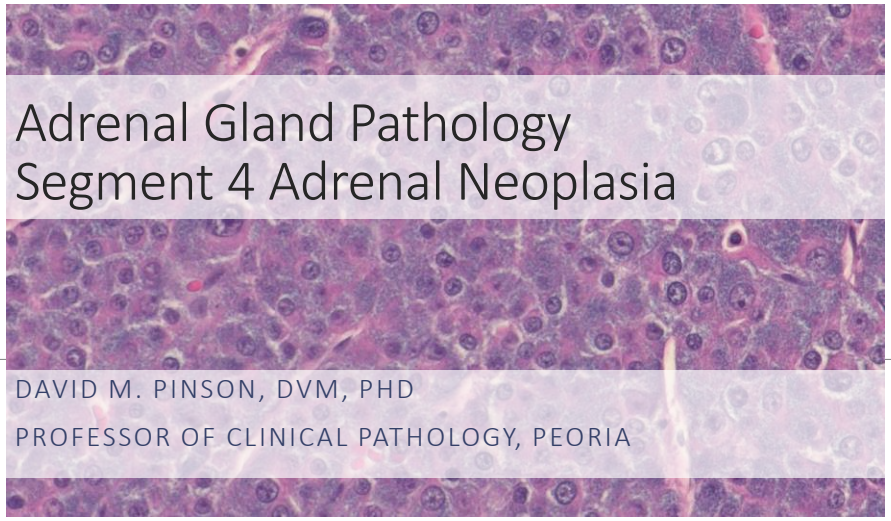


Adrenal Gland Pathology
Segment 4 Adrenal Neoplasia

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CONTENT INTENDED FOR UIC COLLEGE OF MEDICINE STUDENTS ONLY – DO NOT DISTRIBUTE

Greetings students and welcome to the final content recording of this series.

Microlecture Series

Segment	Title	Duration
Segment 1	Overview	4 min 46 sec
Segment 2	Adrenal Hyperfunction	17 min 13 sec
Segment 3	Adrenal Hypofunction	14 min 35 sec
Segment 4	**Adrenal Neoplasia**	**10 min 01 sec**

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This is the fourth recording in a series of four.

Learning Objectives

LO6 Describe the neoplasms of adrenal medulla (pheochromocytoma and neuroblastoma); discuss the pathology of paragangliomas.

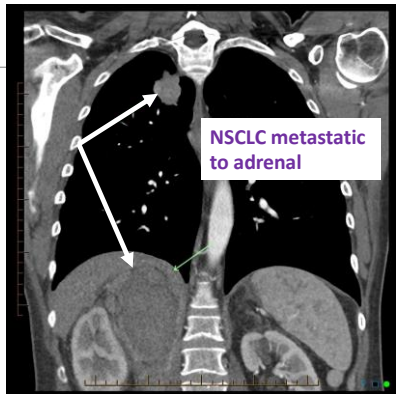
LO7 Describe multiple endocrine neoplasia (MEN) syndrome in terms of clinical, laboratory, and morphologic features as well as prognosis.

In this session, I'll cover the last remaining learning objectives. In addition to the tumors and syndromes in these objectives, I'll also review cortical tumors that may be functional or non-functional.

LO1 LO2 LO3 LO4

Adrenal Tumors

- Primary
  - Cortical
    - Adrenocortical adenoma (ACA)
    - Adrenocortical carcinoma (ACC)
    - Adrenal cyst/pseudocyst
    - Adrenal myelolipoma
  - Medullary
    - Pheochromocytoma
    - Neuroblastoma
- Metastatic: more common than primary



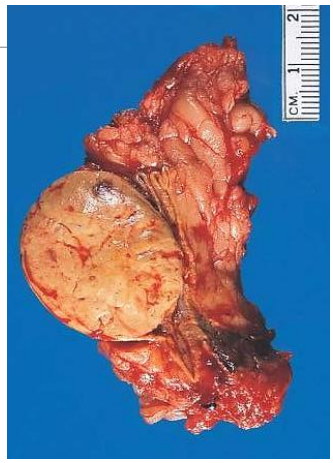
Case courtesy of Dr Chris O'Donnell, Radiopaedia.org, rID: 43135

There are two broad categories of tumors in the adrenal gland. They are tumors primary to the adrenal glands and tumors metastatic to the adrenal glands. The primary tumors can be subdivided into cortical tumors and medullary tumors. Metastatic tumors are more common than primary tumors and the image shows such a metastatic lung tumor in the adrenal gland.

LO1 LO2 LO3 LO4

Adrenal Cortical Tumors

- Morphology does not correlate to function
- Most adenomas are hormonally silent (nonfunctional)
- Most carcinomas are functional (especially in children)
- Frequency
  - Adults: adenoma = carcinoma
  - Children: carcinoma > adenoma
- Most frequent syndromes for functional tumors
  - Adenomas : hyperaldosteronisms; hypercortisolism
  - Carcinomas: virulization; hypercortisolism
- In reality: any cortical neoplasm can produce any cortical hormone



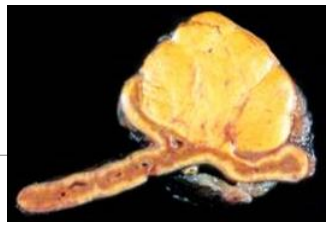
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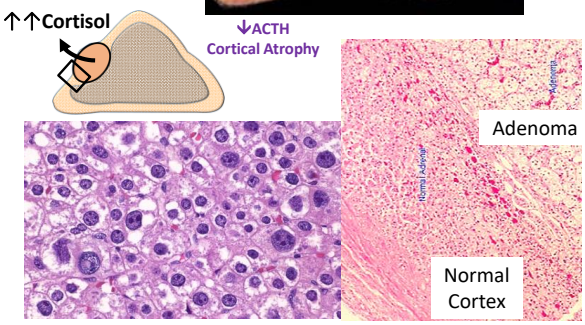
Cortical tumors are usually discrete nodules of variable size. Many of them are yellow from fatty components involved in steroidogenesis. The important take home message is that any cortical tumor can be productive for any hormone. The clinical syndrome likely defines whether or not the tumor is functional. So for cortical tumors, the possibilities include cortisol, aldosterone, or sex steroids and the expected clinical syndromes associated with non-physiologic excess of those hormones. This is what we have been discussing in the second and third recordings in this series.

LO1 LO2 LO3 LO4

Cortical Adenoma



- Morphology
  - <50 g size
  - Circumscribed, encapsulated
  - Often yellow (like fat)
- Status of surrounding adrenal cortex
  - Cortisol producer → atrophy
  - Aldosterone producer → normal
  - Non-productive tumor → normal
- Histology
  - Resembles cortical cells
  - Endocrine atypia
    - Variation in nuclear size
    - Mitoses uncommon



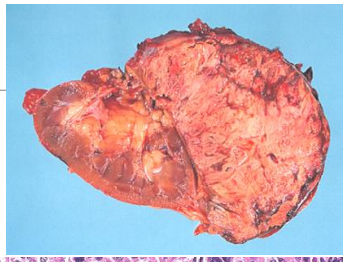
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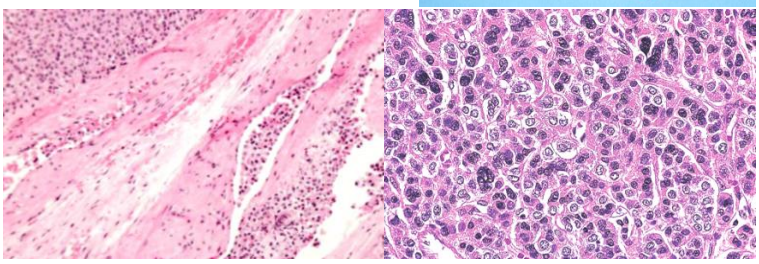
Cortical adenomas are usually small and circumscribed, often less than 50 g. If the tumor is functional, the surrounding adrenal cortex may be profoundly atrophic. As you learned in the second recording ACTH would be very low with tumors producing cortisol and this results in lack of tropic stimulation of the cortex. Tumors producing androgens may result in virulization syndromes. Tumors producing aldosterone would result in hypertension. Non-functional tumors would be clinically silent. They are often identified incidentally.

LO1 LO2
LO3 LO4

Adrenocortical Carcinoma



- Macroscopic Features
  - >100 g; poorly demarcated
  - Necrosis, hemorrhage, cystic degeneration
  - Invasion of adrenal vein, inferior vena cava, lymphatics
  - Metastatic disease: lymph nodes, lungs, liver
- Microscopic features
  - Variable anaplasia
  - Variable mitotic rate



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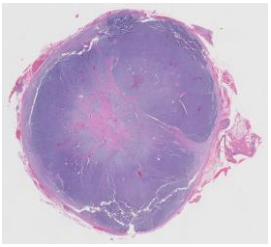
Cortical carcinomas are usually larger, often hemorrhagic, and invasive. This is one of those tumors that likes to invade major veins and even extend up the vena cava. They can metastasize. Like most carcinomas, they can show variable degrees of differentiation and atypia. The images show a large carcinoma invading the kidney and the microscopic images show vascular invasion.

LO6

Adrenal Medullary Neoplasia

Genesis: neural crest cells → sympathogonia → Pheochromoblasts

- Chromaffin cells of adrenal medulla → pheochromocytoma
  - Clinical Features
    - Catecholamine effects (hypertension in 90%)
    - Catecholamine metabolites in urine
      - Vanillylmandelic acid
      - Metanephrines
  - Extra-adrenal chromaffin cells → paraganglioma
- Neuroblasts → neuroblastoma



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For this consideration, the adrenal medulla contains three major cell types. Those are chromaffin cells that produce epinephrine, sustentacular cells in the fibrovascular matrix and neuroblastic cells. The chromaffin cells derive their name from their positive staining with chromium salts and they give rise to the tumor pheochromocytoma. About 90% of these tumors produce epinephrine and a syndrome of hypertension, anxiety, and palpitations. One of the key diagnostic tests is for urinary metabolites of epinephrine. These include metanephrines and villanylmandelic acid or VMA for short. Chromaffin cells are also present in paraganglia of the autonomic nervous system and can give rise to tumors at other sites. I'll discuss neuroblastoma in a few slides.

LO6

Pheochromocytoma

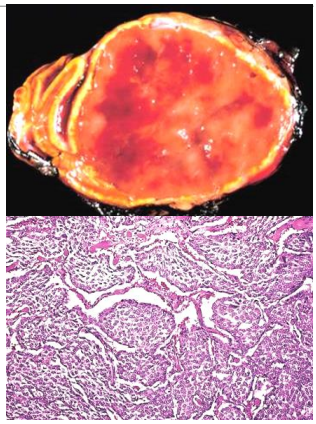
- The "10%" tumor
  - 10% of pheochromocytomas are extra-adrenal (paragangliomas)
  - 10% of sporadic adrenal pheochromocytomas are bilateral (up to 50% of familial adrenal pheochromocytomas are bilateral)
  - 10% of sporadic adrenal pheochromocytomas are malignant (20-40% of paragangliomas are malignant)
  - 10% of adrenal pheochromocytomas are found in children
  - 10% of sporadic adrenal pheochromocytomas are not associated with hypertension
  - Exception to "10% rule": up to 25% of pheochromocytomas and paragangliomas harbor germline mutations, and are thus familial.

Pheochromocytoma is known as the 10% tumor for all of the reasons listed. These features define much of the clinical syndrome. The notable exception is that 25% may be familial and associated with the multiple endocrine neoplasia or MEN syndromes.

LO6

Pheochromocytoma

- Gross Morphology
  - 1 - 4000 g (avg. 100 g)
  - Well-demarcated; yellow-tan ("dusky") to red
  - Hemorrhagic, necrotic, cystic
- Microscopic Morphology
  - "Zellballen" pattern: neuroendocrine packaging in light vascular stroma
  - Neuroendocrine markers
    - Synaptophysin
    - Chromogranin
    - Neuron specific enolase
    - S100 (sustentacular cells) – may be absent in malignant tumors



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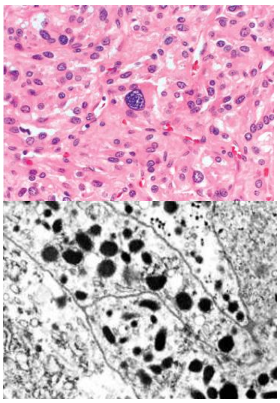
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Pheochromocytoma is a neuroendocrine type tumor. The classical histologic pattern is termed zellballen. This is a unique pattern of cells in packets supported by delicate fibrovascular stroma. The sustentacular cells are typically found in the stroma, but may not be present in malignant tumors. These sustentacular cells are usually positive for the S100 marker. Most neuroendocrine type tumors may be positive for chromogranin, synaptophysin, or neuron specific enolase.

LO6

Pheochromocytoma

- Criteria for malignancy
  - **definite: metastases**
    - lymph nodes
    - liver
    - lungs
    - bone
  - other criteria associated with aggressive behavior
    - high mitotic count/atypical forms
    - spindle cell morphology
    - confluent tumor necrosis



NEUROSECRETORY GRANULES

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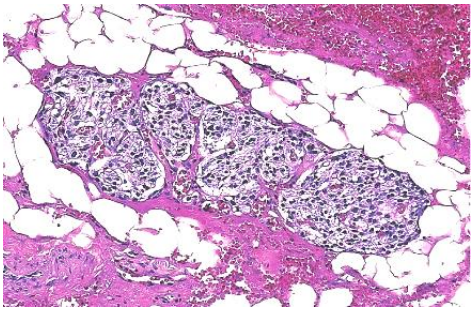
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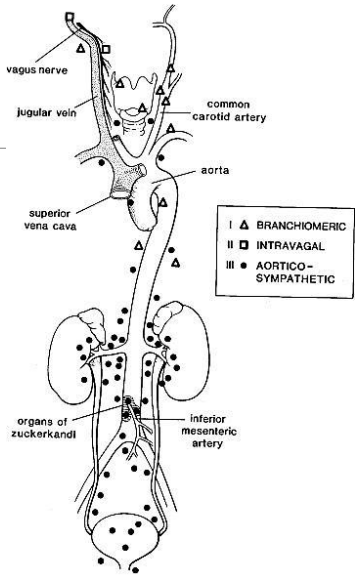
Pheochromocytoma cells show variable degrees of atypia and contain neurosecretory granules. One of the challenges in pheochromocytoma diagnosis is determining whether or not the tumor is malignant. The morphology does not provide very many clues. The cells can be malignant and not show that much atypia. The absolute criterion for malignancy is whether or not it metastasizes. High mitotic count or necrosis might offer clues but these features are not absolute and totally reliable. Metastasis is the golden criterion.

LO6

Paragangliomas

- Extra-adrenal pheochromocytomas
- Sites: any organ containing paraganglionic tissue





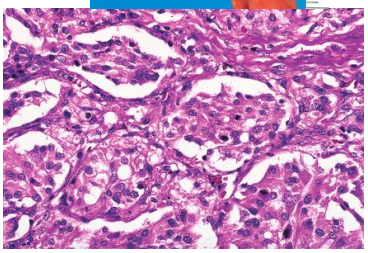
Pheochromocytomas outside of the adrenal gland are called paragangliomas. They are also neuroendocrine tumors. They can occur in any tissue with a ganglion. The right image shows the distribution of such ganglia and the possible sites for tumor.

LO6

Paragangliomas

- Common locations
  - Paravertebral: sympathetic
    - Organs of Zuckerkandl
    - Urinary bladder
  - Aorticopulmonary (head/neck): parasympathetic: ~ 70%)
    - Carotid body(carotid body tumor)
    - Jugulotympanic body (glomus jugulare tumor)
- Malignancy
  - Head/neck: 10%
  - Retroperitoneum: 40%





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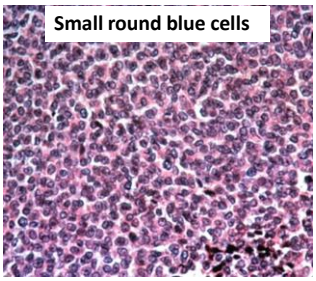
The most common paraganglioma locations are paravertebral and aorticopulmonary. The zellballen pattern is consistent at these sites. The majority of these tumors are benign, but some are malignant…moreso at the retroperitoneal sites.

LO6

The "Blastomas"

Small round blue cell tumors of childhood

- Differential diagnosis
  - –Neuroblastic tumors
    - **Neuroblastoma (adrenal medulla)**
    - Retinoblastoma
    - Medulloblastoma
  - –Wilms tumor (kidney)
  - –Lymphoma
  - –Rhabdomyosarcoma
  - –Ewing sarcoma/PNET
  - –Leukemia
- Definitive diagnostic methods
  - –Immunohistochemistry
  - –Electron microscopy
  - –Chromosome analysis



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The next tumor is neuroblastoma. This is one of the small round blue cell tumors of childhood. The differential diagnosis for small round blue cell tumors is listed on this slide. The image illustrates this namesake. Most of these tumors are composed of nondescript infiltrates of round cells supported by scant stroma. The diagnosis is often made or confirmed with immunohistochemistry and chromosome analysis.

LO6

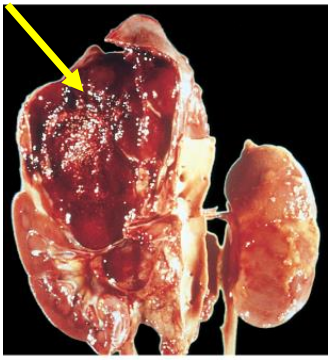
Neuroblastoma

Most common extracranial solid tumor of childhood

Frequency: 1 in 7000 live births

Locations
- adrenal medulla (40%)
- Abdominal paravertebral (25%)
- Posterior mediastinum 15%)
- Sympathetic chain (20%)

Gross morphology
- Soft, gray-tan
- Necrosis, hemorrhage



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Neuroblastoma is the most common extra-cranial solid tumor of childhood. You will recall that lymphomas and leukemias are the most common tumors of childhood, but they are not solid tumors except when they occur in lymph nodes. Neuroblastoma is fairly common. The most common locations are indicated and you can see why we put this in the adrenal gland lecture. The image shows a neuroblastoma effacing half of the kidney. Of course a gross differential for this image might be Wilms tumors.

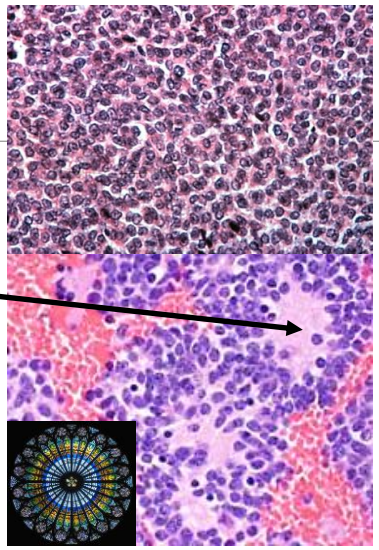
LO6

Neuroblastoma: histology

Small-, round-, blue-cell tumor of neuroblasts

Anaplasia: variable

Homer-Wright pseudorosettes: surround areas of neuropil.



UNFAVORABLE HISTOLOGY (BOTH)

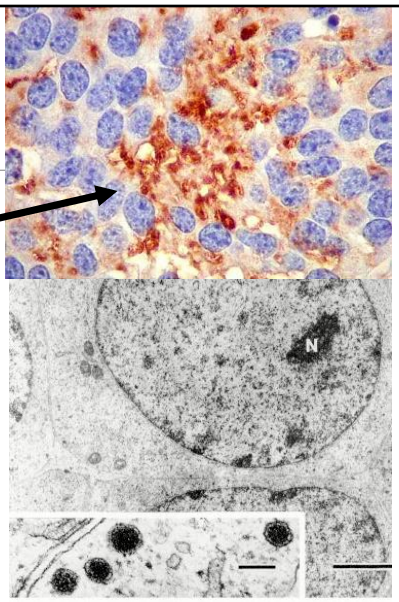
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As I suggested, neuroblastoma is a small round blue cell tumor, but it can have some unique patterns. One of those is Homer-Wright pseudorosettes. These are cells oriented around neuropil. They are not true rosettes, which only occur in ependymomas of the CNS. The term rosettes comes from references to this pattern in windows in European gothic churches.

LO6

Neuroblastoma: neuroendocrine differentiation

- Immunohistochemistry: neuroendocrine markers
- Electron microscopy: neurosecretory granules
- Tumor markers
  - Blood: catecholamines
  - Urine catecholamine metabolites
    - Vanillylmandelic acid (VMA)
    - Homovanillic acid (HVA)



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Neuroblastoma is also a neuroendocrine tumor with secretory granules. They react to chromogranin and synaptophysin stains. They can produce catecholamines that are metabolized and excreted in the urine.

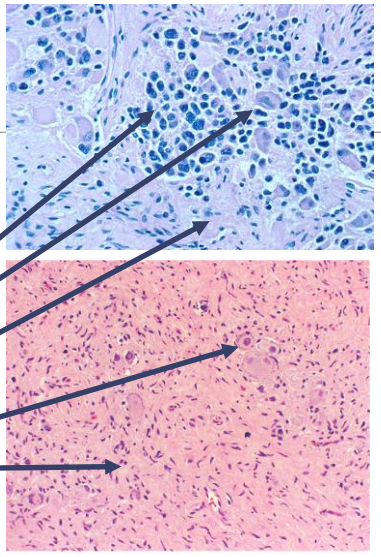
LO6

Neuroblastoma: differentiation, regression

Schwannian stroma = favorable histology

Ganglioneuroblastoma
- Neuroblasts
- Ganglion cells
- Schwannian stroma

Ganglioneuroma (benign)
- Ganglion cells
- Schwannian stroma



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These tumors can be more differentiated toward ganglion morphology. So…there are two additional diagnoses that can be made based on the degree of differentiation. They Ganglioneuroblastoma and Ganglioneuroma. These tumors have variable degrees of what is called Schwannian stroma, a prognostically favorable feature.

LO6

Prognostic Indicators for Neuroblastoma

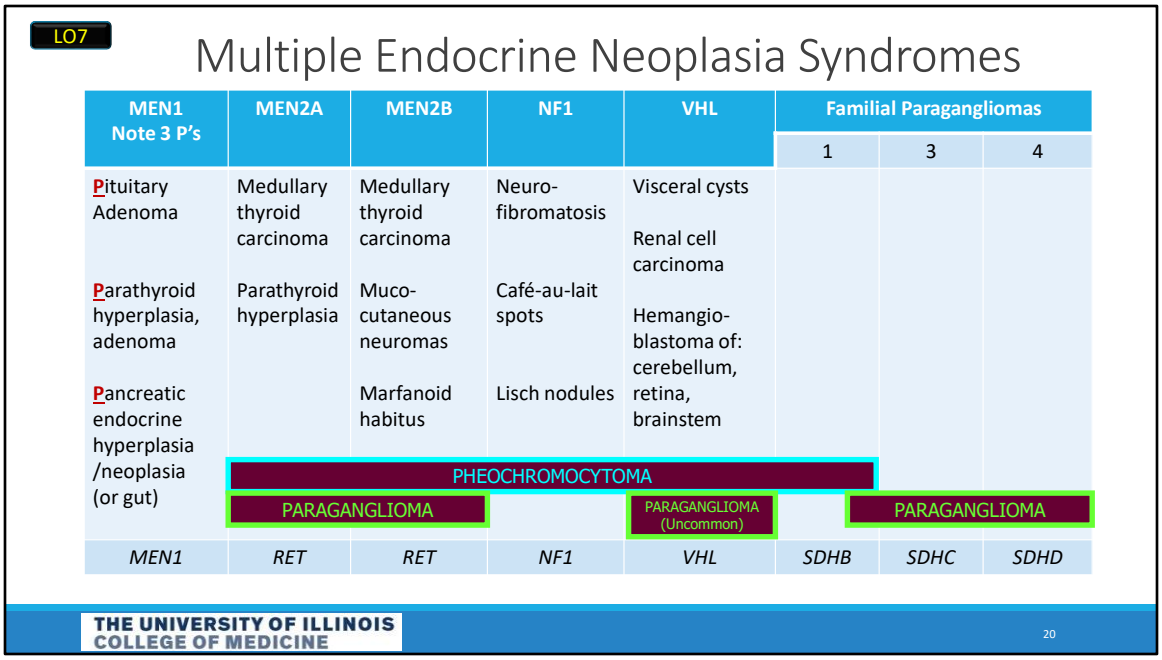
- Favorable prognosis
  - **Stage 1, 2A, 2B, 4S**
  - **Age <18 months**
  - Schwannian stroma
  - Low mitotic index
  - **Hyperdiploid state**
- Unfavorable prognosis
  - **Stage 3, 4**
  - **Age >18 months**
  - No Schwannian stroma
  - High mitotic index
  - **Near diploid state**
  - **Myc amplification**

Clinical Staging for Neuroblastoma

Stage	Features
Stage 1	• **Localized, complete excision**
Stage 2A	• **Localized, incomplete excision** • **Ipsilateral nodes negative**
Stage 2B	• **Localized, incomplete excision** • **Ipsilateral nodes positive** • **Contralateral nodes enlarged , but negative**
Stage 3	• **Unresectable** • **Tumor crosses midline** • **Contralateral nodes positive**
Stage 4	• **Primary tumor with metastasis to distant nodes, bone, bone marrow, skin, other organs**
Stage 4S	• **Infants <1 year** • **Localized primary** • **Skin, liver, and/or bone marrow**

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There are a collection of prognostic indicators for neuroblastoma and those are detailed on this slide. The ones indicated in red font are the most defining for favorable and unfavorable prognosis. Whenever it is present, myc amplification makes the tumor high risk. This is USMLE board type information.



Approximately 25% of pheochromocytomas are familial and many of them are encountered in the multiple endocrine neoplasia syndromes. Those syndromes are summarized on this slide.

MEN1 syndrome is associated with a mutation in the MEN gene and is the syndrome of 3 P's. Tumors can develop in the pituitary, the parathyroid, and classically the pancreas. However, some of these tumors may develop in the gastrointestinal tract as carcinoids and still be part of MEN1 syndrome, especially if pituitary and parathyroid are involved.

The MEN2 syndromes include pheochromocytoma. One of the important tumors in MEN2 is medullary thyroid carcinoma. The development of this tumor is so consistent that thyroidectomy is often recommended for these patients. MEN2 can also have parathyroid proliferative lesions as well paragangliomas. MEN2A and MEN2B are slightly different in clinical presentation. The MEN2 syndromes are associated with RET mutations.

Other genetic syndromes associated with pheochromocytoma include neurofibromatosis and Von Hippel Lindau.

Summary

- Summarized the major tumors of the adrenal cortex, including functional and non-functional adenomas and carcinomas.
- Summarized the tumor of the adrenal medulla, including pheochromocytoma and related paragangliomas.
- Summarized the clinical, histologic and prognostic features of neuroblastoma.
- Summarized the tumor types associated with multiple endocrine neoplasia syndromes and syndromes that include pheochromocytoma.
- This concludes the series on adrenal gland pathology.

In this recording, I have summarized the major cortical tumors and medullary tumors. I described paragangliomas as similar to pheocytomocytoma at other sites. I described the differential diagnosis of small round blue cell tumors of childhood and summarized details about neuroblastoma. I highlighted pheochromocytoma and medullary thyroid carcinoma in the multiple endocrine neoplasia syndromes. This concludes the series on adrenal gland pathology.